॥वसुधैव कुटुम्बकम्॥

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE - Academic Year 2026-27

Java Programming – Lab (Batch 2025-29)

Lab Assignment No:- 21

Name of Student	Abobakr Mohamed Elameen
PRN No.	25070122015
Batch	2025-29
Class	Programming with JAVA
Academic Year & Semester	2nd Year – Sem III
Title of Assignment	Demonstration and exploration of JDBC classes and interfaces (DriverManager, Connection, Statement).
Practice questions	1. Create a program to connect Java with database and display database connection status.

```
package JAVA_EXP_21;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class ConnectionStatusDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/testdb";
        String user = "root";
        String password = "XXXXX";

        try (Connection conn = DriverManager.getConnection(url,
user, password)) {
            if (conn != null && !conn.isClosed()) {
                System.out.println("Connection status:
Connected");
            }
        } catch (SQLException e) {
```

```
            System.out.println("Connection status: Failed - " +
e.getMessage());
        }
    }
}
```

2. Create a Java program to connect with a database and display the connection status. After successful connection, show a message indicating that the Student database is connected successfully.

```
package JAVA_EXP_21;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class StudentDatabaseConnectionDemo {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/testdb";
        String user = "root";
        String password = "XXXXX";

        try (Connection conn = DriverManager.getConnection(url,
user, password)) {
            System.out.println("Connection status: Connected");
            System.out.println("Student database is connected
successfully");
        } catch (SQLException e) {
            System.out.println("Connection status: Failed - " +
e.getMessage());
        }
    }
}
```

Output

1)

```
C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS>java -cp .
onStatusDemo
Connection status: Connected

C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS>
```

	2) C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS>java -cp .;JA atabaseConnectionDemo Connection status: Connected Student database is connected successfully C:\Users\bkor\OneDrive\Desktop\JAVA_PRACTICALS>
Conclusion	DriverManager creates the link to the database, Connection represents the live session and provides access to metadata and statements, and Statement executes SQL through that session. Together they form the foundation of every JDBC application. Confirming a successful connection is the first step before running any queries or updates.